

Experiment 15:

Implement Iris Flower Classification using Naive Bayes classifier

Python Code:

```
from sklearn.datasets import load_iris
from sklearn.naive_bayes import GaussianNB
iris = load_iris()
X, y = iris.data, iris.target
model = GaussianNB()
model.fit(X, y)
sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(sample)[0]
print("Sample:", sample)
print("Predicted Iris Flower:", iris.target_names[prediction])
```

Output:

```
Sample: [[5.1, 3.5, 1.4, 0.2]]
Predicted Iris Flower: setosa
```